

Qartu Books Resources for Schools

AS 91261 Apply algebraic methods in solving problems

L2 Algebra

10 Revision Sheets

Covers mainly Achieved and Merit questions

Approx 10 questions in each revision sheet

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1. Expand: $(x + 8)(5 - x)^2$

6. Solve by factorizing: $x^2 + 9x = 22$

2. Factorise: $3x^2 + 7x + 2$

7. Solve by completing the square:
 $x^2 + 12x + 20 = 0$

3. Simplify. Give answer with positive indices:
 $\left(\frac{27x^6}{64y^{-3}}\right)^{\frac{2}{3}}$

8. Find k if the equation $kx^2 - 3x - 2 = 0$
has one repeated root.

4. Simplify: $2\log(x) - \log(2x) + \log(4x)$

9. Solve: $4^{x+1} = 19$

5. Simplify: $\frac{3y}{10} - \frac{y+3}{15}$

10. Solve: $\log_x(93) = 5$

1. Expand: $(2x - 7)^2$

6. Solve: $\frac{2x-3}{3} + \frac{x+4}{2} \leq \frac{x}{12}$

2. Factorise: $8x^2 - 32$

7. Solve by completing the square:
 $5x^2 + 20x + 1 = 0$

3. Simplify. Give answer with positive indices:
 $\frac{(2p^{-2}q)^{-2}}{(2pq)^3}$

8. Find k if the equation $4x^2 + kx + 9 = 0$ has two real roots.

4. Simplify: $\log(xy) - 2\log(y)$

9. Solve: $27^{1-2x} = \frac{1}{2}$

5. Simplify: $\frac{(x-3)(x+2)}{6x} \times \frac{3x^2}{7(x+2)}$

10. Solve: $\frac{1}{3}\log_2(2x) = 4$

1. Expand: $(x + 1)(2x - 5)^2$

6. a) Simplify: $\frac{1}{3}\log_3(27) + \log_3(3)$

b) Solve: $2^{x+1} = 39$

2. Factorise:

a) $x^2 - 8x - 20$

7. Solve using algebra:
 $2x^2 - 5x = -3$

b) $5x^2 + 19x - 4$

c) $4x^2 - 4$

3. Simplify. Give answer with positive indices:

$$\left(\frac{2y}{x^2}\right)^{-2}$$

8. Solve: $\frac{y-8}{3} < 2y - 1$

4. Make x the subject:

$$y = \frac{1 - 2x}{4x - 3}$$

9. Show that $y = 3x - 4$ is tangent to
 $y = x^2 - x$

10. Simplify: $\frac{2x^2+9x-5}{4x-2}$

5. Simplify:

a) $\frac{3}{xy} + \frac{2}{5}$

11. Find equation of parabola which has
 x -intercepts of -4 and 3 and y -intercept of -12 .

b) $\frac{3b}{4d} \times \frac{2d}{5c}$

1. Expand: $(x + 2)(x + 4)(x + 5)$

6. a) Simplify: $\frac{5}{2}\log(4) + \frac{1}{2}\log(64)$

2. Factorise:

a) $x^2 - 27x + 140$

b) $3x^2 + 20x - 7$

c) $5x^2 + 30x$

b) Solve: $7^{2x+3} = 60$

7. Solve using algebra:
 $3x^2 - 14x = 5$

3. Simplify. Give answer with positive indices:

$$\left(2x^{-\frac{1}{2}}\right)^4$$

4. Make x the subject:

$$y = \sqrt{\frac{x-5}{x+3}}$$

8. Find k if the roots of the equation

$$2x^2 - 9x + k = 0 \quad \text{are} \quad \frac{m}{2} \quad \text{and} \quad m - 3;$$

5. Simplify:

a) $\frac{32x}{15yz} \div 4x$

9. Solve: $\frac{5x-2}{3} = 4(x-1)$

10. $y = x^2 - x - 2$; for what values of x is the function positive?

b) $\frac{3}{x-1} + \frac{2}{x+5}$

1. Expand: $(x - 6)^3$

6. a) Simplify $\frac{2}{3}\log(27) + \frac{4}{3}\log(64)$

2. Factorise:

b) Solve: $5^x = 3$

a) $x^2 + 5x - 104$

b) $4x^2 + 16x - 48$

7. Solve using algebra:
 $3x^2 - 7x = -4$

c) $25 - 9x^2$

3. Simplify. Give answer with positive indices:

$$\left(3x^{\frac{3}{2}}y^{\frac{3}{4}}\right)^{-2}$$

8. Solve: $\frac{x+4}{2} < 3(x-1)$

4. Make x the subject:

$$y = (2x - 1)^2$$

9. Simplify: $\frac{x^2 + 3x}{x^2 + 6x + 9}$

5. Simplify:

a) $\frac{2}{x} - \frac{3}{x+2}$

10. Solve: $\frac{3(2x+1)}{5} - \frac{5x+1}{4} = \frac{1}{2}$

11. Find the equation of a parabola with the vertex at $(2, 3)$ and y -intercept of 5.

b) $3x \div \frac{4y}{5x}$

1. Expand: $(3x^2 + 7)(3x^2 - 7)$

6. a) Simplify $2\log(3) - 3\log(2)$

2. Factorise:

a) $x^2 - 3x - 18$

b) $72 + 18x - 2x^2$

b) Evaluate: $\log_3(5)$

c) $10x^2 - 40y^2$

7. Solve using algebra:

$$2x^2 - 13x = 7$$

3. Simplify. Give answer in index form:

$$\frac{7}{4\sqrt[3]{x^5}}$$

8. Find the value of a if $ax^2 + 6x + 8 = 0$ has no real roots.

4. Make x the subject:

$$y - 5 = \frac{2x+1}{x-1}$$

9. Simplify: $\frac{x+1}{2(x-2)} - \frac{x+3}{4(x-3)}$

5. Simplify:

a) $\frac{xy}{2} \times \frac{6}{2x^2} \times \frac{(3x)^2}{9x}$

10. Find the equation of parabola with the vertex at $(-3, -5)$ and y -intercept of -2.75 .

b) $\frac{5}{x+1} - \frac{x+1}{3}$

1. Expand: $(4x^2 + 5y^2)^2$

6. a) Simplify $\frac{3\log(2)+\log(4)}{\log(8)}$

2. Factorise:

a) $x^2 - 52x + 640$

b) Evaluate: $\log_2(6)$

b) $20x^2 + 11x - 4$

7. Solve using algebra:
 $3 + 5x - x^2 = 0$

c) $3x - 12x^3$

3. Simplify. Give answer in index form:

$$\left(27a^{\frac{3}{4}}\right)^{\frac{1}{3}} \times \left(9a^{\frac{-3}{2}}\right)^{\frac{-1}{2}}$$

8. Find the value of k if
 $(2k + 3)x^2 - 4kx + 4 = 0$ has equal roots.

4. Make x the subject:

$$2y = \sqrt{x+3} - 2$$

9. Simplify: $\frac{2x^2-3x-2}{x^2-4}$

5. Simplify:

a) $\left(\frac{10}{x^2}\right)^2 \times \left(\frac{x^3}{5}\right)^2$

10. Find the equation of parabola with the x -intercepts of -1 and 2 , and y -intercept of 0.6 .

b) $\frac{x}{x-4} - \frac{3}{x+5}$

1. a) Solve: $\log(x) = 3\log(2)$

6. Find k if $(2x - 3)(x + 4) = k$ has only one real solution.

b) Solve: $\log_5(x) = 2$

2. $9^n - 6 \times 3^n - 27 = 0$ has only one real root. Explain why.
(Hint: $3^n = x$)

7. Given: $A = 2000(1.035)^t$

Find t if $A = 2250$.

3. a) Factorise: $5x^2 - 9x - 2$

8. Solve: $\frac{x^2 - 5x + 6}{x^2 + x - 6} = 4$

b) Solve: $5x^2 - 9x - 2 = 0$

c) Solve: $\frac{x^2 + 5x + 2}{x + 2} = 3$

9. Find c if $\frac{x^2 + x - 6}{6x^2 + 4x + c} = \frac{x + 3}{2(3x + 8)}$

4. Simplify:

a) $(x^5)^2(2x)^3$

b) $\left(8x^{\frac{1}{2}}\right)^{\frac{2}{3}}$

10. Simplify: $\sqrt{\frac{\left(8x^{\frac{1}{2}}\right)^{\frac{2}{3}}}{x^{\frac{-1}{2}}}}$

5. Solve by using the quadratic formula:

$(2x - 3)(x + 4) = 13$

1. a) Factorise: $6x^2 - 11x - 10$

6. Make x the subject: $a^x = 5^{(x-1)}$

b) Solve: $6x^2 - 11x - 10 = 0$

7. One root of the equation $3x^2 + 4x - k = 0$ is $x = 2$. Find k and the second root.

2. Find m if $4x^2 - 8x + m = 0$ has only one real root.

8. Solve: a) $\log_x(64) = 3$

3. Simplify: $\frac{2x^2 - 8}{x^2 - 2x - 8}$

b) $\frac{3 \times 2^{x+1}}{8^x} = 96$

4. Solve for x , if

$$(x + 2) - 3\sqrt{x + 2} - 4 = 0$$

Has only one real solution.

(Hint: use notation $a = \sqrt{x + 2}$)

9. Equation $(3x + 1)^2 = -7$ has no real solutions. Explain why, graphically and algebraically.

5. Simplify:

a) $\frac{(4a^2)^3}{(8a^5)^2}$

b) $(16x^2)^{\frac{1}{4}}$

c) $(16x^2)^{\frac{1}{4}} \times (9x^3)^{\frac{1}{2}}$

10. Solve for x in terms of m :

$$\log(x) = 2\log(mx)$$

1. Simplify:

a) $\left(\frac{5}{a^4}\right)^{-3}$

b) $(0.25x^3)^{\frac{1}{2}}$

c) $\frac{(8x^6)^{\frac{1}{3}}}{3(x^{-2})^4}$

2. Find m if one root of $x^2 + mx + 12 = 0$ is three times the other root.3. Find n if $3x^2 - nx + 5 = 0$ has 2 different real roots.

4. Solve by factorising:

$$10x^4 - 13x^2 + 4 = 0$$

5. Solve by factorising:

$$12a^2 - 11a - 15 = 0$$

6. Simplify: $\frac{3}{x-2} - \frac{4x}{x+1}$

7. Solve: $\frac{x^2+2x-8}{x^2-x-2} = 3$

8. Make x the subject:

$$h = rx^2 - tx$$

9. Find the equation of a parabola if the maximum value of y is 6 and x -intercepts are 0 and 12.10. Solve for x :

a) $x = \log_3(81)$

b) $\log_x(343) = 3$

c) $5^x \times 2^{-2x} = 15$

Algebra Revision 1 - Answers:

1. $x^3 - 2x^2 - 55x + 200$

2. $(3x + 1)(x + 2)$

3. $9x^4y^2/16$

4. $\text{Log}(2x^2)$

5. $(7y - 6)/30$

6. $x = -11 \text{ or } x = 2$

7. $x = -2 \text{ or } x = -10$

8. $k = -9/8 = -1.125$

9. $x = 1.124$

10. $x = 2.476$

Algebra Revision 2 - Answers:

1. $4x^2 - 28x + 49$

2. $8(x - 2)(x + 2)$

3. $p/32q^5$

4. $\log(x/y)$

5. $x(x - 3)/14$

6. $x \leq -12/13$

7. $x = -0.051 \text{ or } x = -3.949$

8. $k < -12 \text{ or } k > 12$

9. $x = 0.605$

10. $x = 2048$

Algebra Revision 3 - Answers:

1. $4x^3 - 16x^2 + 5x + 25$
2. a) $(x - 10)(x + 2)$
b) $(x - 4)(5x - 1)$
c) $4(x - 1)(x - 1)$
3. $x^4/4y^2$
4. $x = (1 + 3y)/(4y + 2)$
5. a) $(15 + 2xy)/5xy$
b) $3b/10c$
6. a) 2
b) 4.285
7. $x = 1$ or $x = 3/2$
8. $y > -1$
9. $(x - 2)^2 = 0$, so $x = 2$ only one root
10. $(x + 5)/2$
11. $y = (x + 4)(x - 3)$ or equivalent

Algebra Revision 4 - Answers:

1. $x^3 + 11x^2 + 38x + 40$
2. a) $(x - 7)(x - 20)$
b) $(3x - 1)(x + 7)$
c) $5x(x + 6)$
3. $16/x^2$
4. $x = (3y^2 + 5)/(1 - y^2)$
5. a) $8/15yz$
b) $(5x + 13)/(x^2 + 4x - 5)$
6. a) $\log(4)$
b) -0.4480
7. $x = 5$ or $x = -1/3$
8. $k = 10, m = 5$
9. $x = 10/7$
10. $x < -1, x > 2$

Algebra Revision 5 - Answers:

1. $x^3 - 18x^2 + 108x - 216$
2. a) $(x + 13)(x - 8)$
b) $4(x + 6)(x - 2)$
c) $(5 - 3x)(5 + 3x)$
3. $1/9x^3y^{3/2}$
4. $x = \frac{1 \pm \sqrt{y}}{2}$
5. a) $(-3x+4)/(x^2+2x)$
b) $(15x^2)/(4y)$
6. a) $\log(2304)$
b) 0.6826
7. $x = -4/3$ or $x = -1$
8. $x > 2$
9. $x/(x + 3)$
10. $x = -3$
11. $y = 0.5(x - 2)^2 + 3$

Algebra Revision 6 - Answers:

1. $9x^4 - 49$
2. a) $(x - 6)(x + 3)$
b) $2(12 - x)(x + 3)$
c) $10(x - 2y)(x + 2y)$
3. $(7/4)x^{-5/3}$
4. $x = \frac{y-4}{y-7}$
5. a) $3y/2$
b) $(14 - 2x - x^2)/(3x+3)$
6. a) $\log(9/8)$
b) 1.465
7. $x = 7$ or $x = -0.5$
8. $a > 1.125$
9. $(x^2 - 5x)/4(x - 2)(x - 3)$
10. $y = 0.25(x + 3)^2 - 5$

Algebra Revision 7 - Answers:

1. $16x^4 + 40x^2y^2 + 25y^4$
2. a) $(x - 32)(x - 20)$
b) $(5x + 4)(4x - 1)$
c) $3x(1 - 2x)(1 + 2x)$
3. a
4. $x = 4y^2 + 8y + 1$
5. a) $4x^2$
b) $(x^2 + 2x + 12)/(x^2 + x - 20)$
6. a) $5/3$
b) 2.585
7. $x = -0.54$ or $x = 5.54$
8. $k = 3$ or $k = -1$
9. $(2x + 1)/(x + 2)$
10. $y = -0.3(x + 1)(x - 2)$

Algebra Revision 8 - Answers:

1. a) $x = 8$
b) $x = 25$
2. $3^n = 9$, so $n = 2$ (as 3^n cannot be negative)
3. a) $(5x + 1)(x - 2)$
b) $x = -1/5$ or $x = 2$
c) $x = 1.236$ or $x = -3.236$
4. a) $8x^{13}$
b) $4x^{1/3}$
5. $x = 2.5$ or $x = -5$
6. $k = -15.125$
7. $t = 3.42$
8. $x = -5$ (note that $x = 2$ is not a valid solution)
9. $c = -32$
10. $2x^{5/12}$

Algebra Revision 9 – Answers:

1. a) $(2x - 5)(3x + 2)$
b) $x = -2/3, \quad x = 5/2$
2. $m = 4$
3. $2(x - 2)/(x - 4)$
4. $x = 14, \quad (x + 2)^{0.5} = -1$ is not possible, as surd can't be negative.
5. a) $1/a^4$
b) $2x^{1/2}$
c) $6x^2$
6. $x = -\log(5) / (\log(a) - \log(5))$
7. $k = 20$
8. a) $x = 4$
b) $x = -2$
9. *Explanation and graph*
10. $x = 1/m^2$

Algebra Revision 10 – Answers:

1. a) $a^{12} / 125$
b) $0.5x^{3/2}$
c) $(2/3)x^{10}$
2. $m = \pm 8$
3. $n > 7.7 \quad \text{or} \quad n < -7.7$
4. $x = \pm 0.894, \quad x = \pm 0.707$
5. $a = 5/3 \quad \text{or} \quad a = -3/4$
6. $(-4x^2 + 11x + 3)/(x^2 - x - 2)$
7. $x = 0.5 \quad \text{and} \quad x \neq 2$
8. $x = (t \pm \sqrt{t^2 + 4rh}) / 2r$
9. $y = (-1/6)x^2 + 2x \quad \text{or} \quad y = (-1/6)x(x - 12)$
10. a) $x = 4$
b) $x = 7$
c) $x = 12.14$